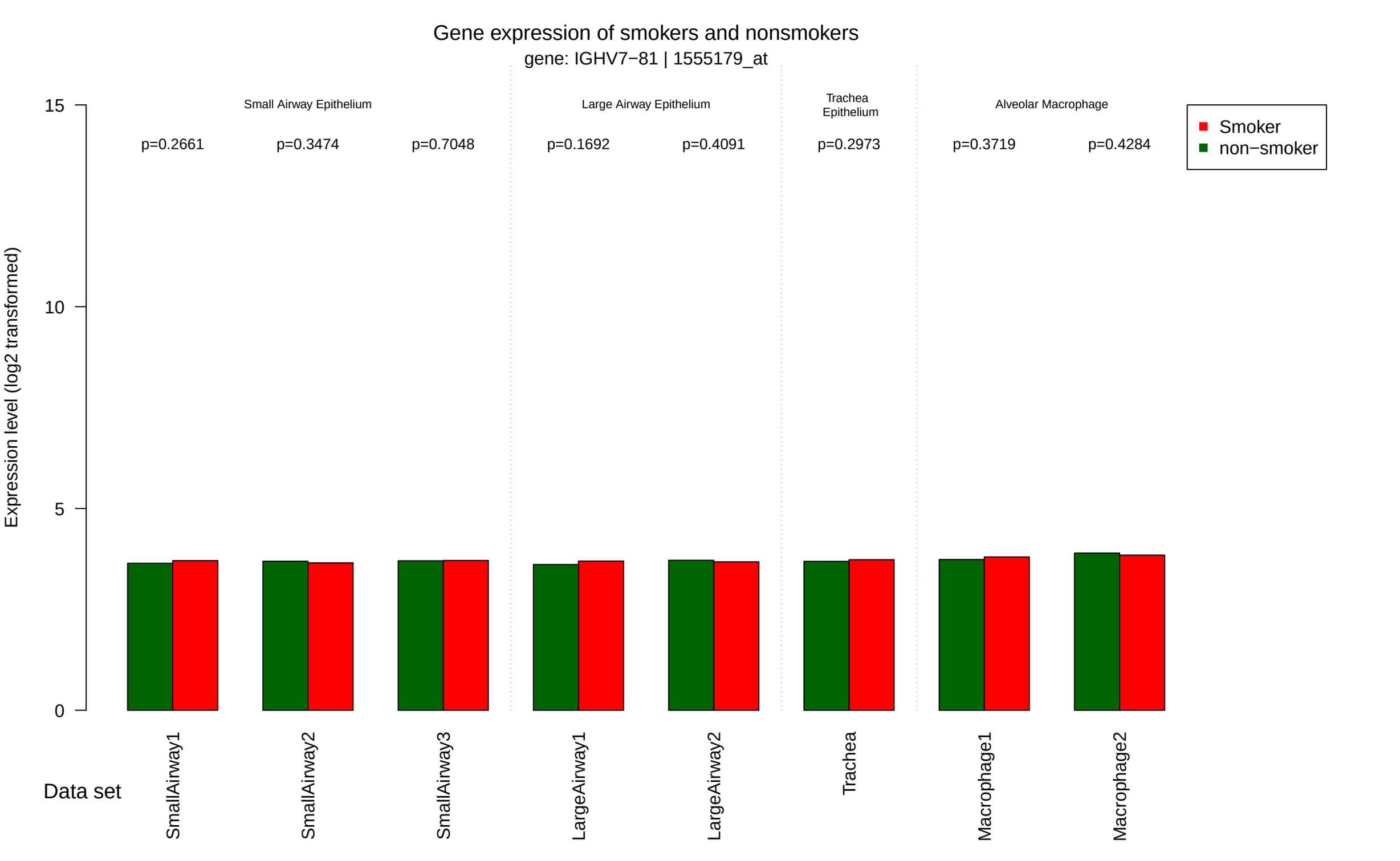
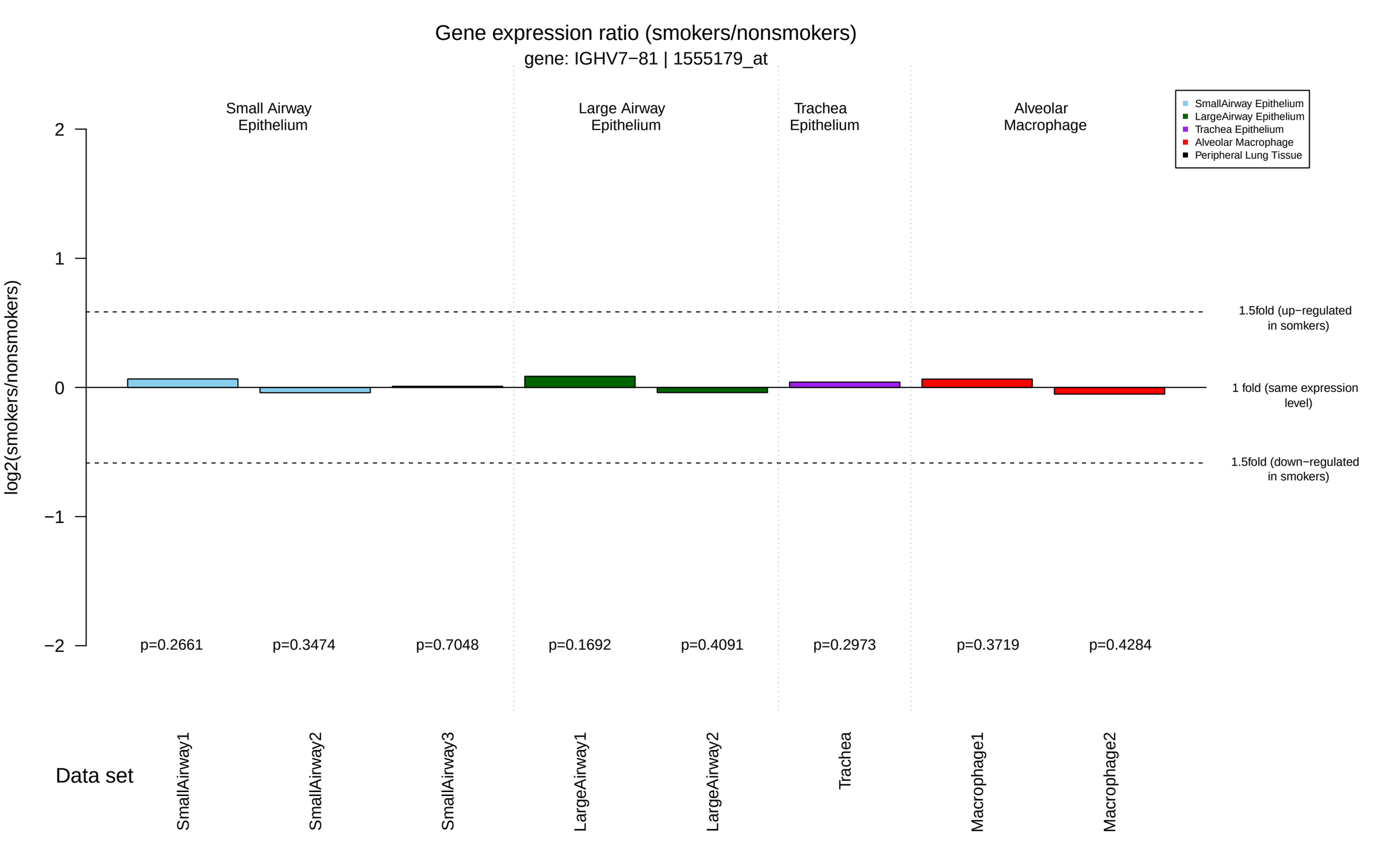
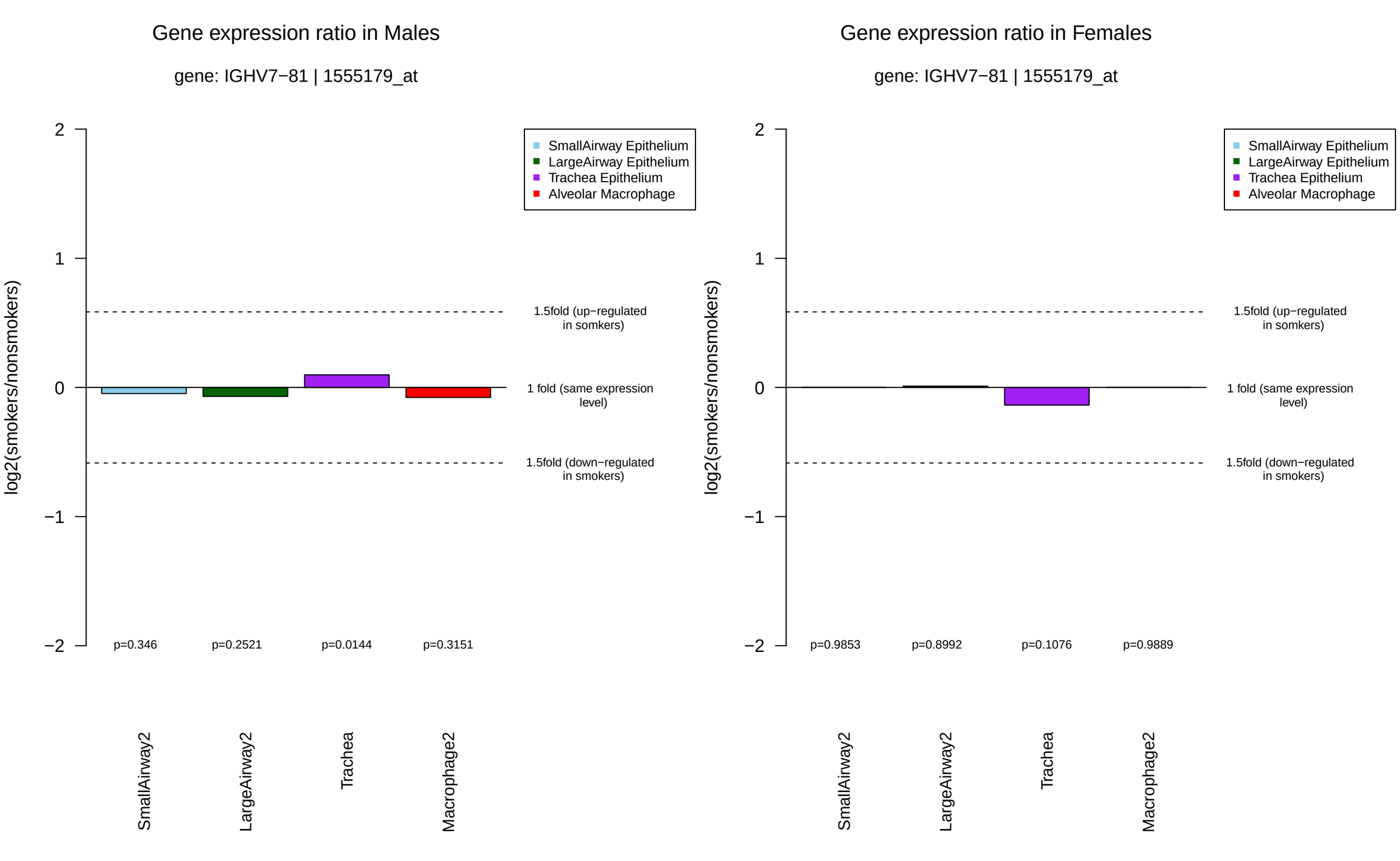
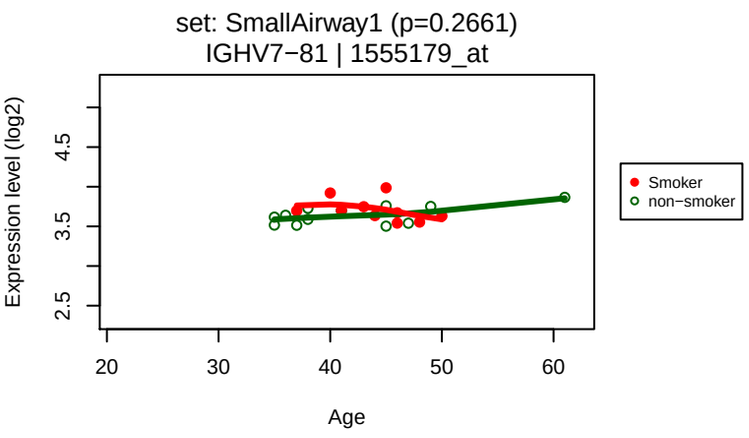
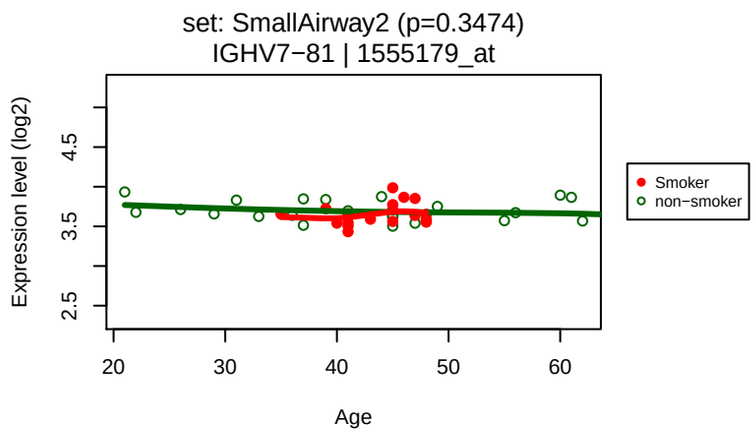
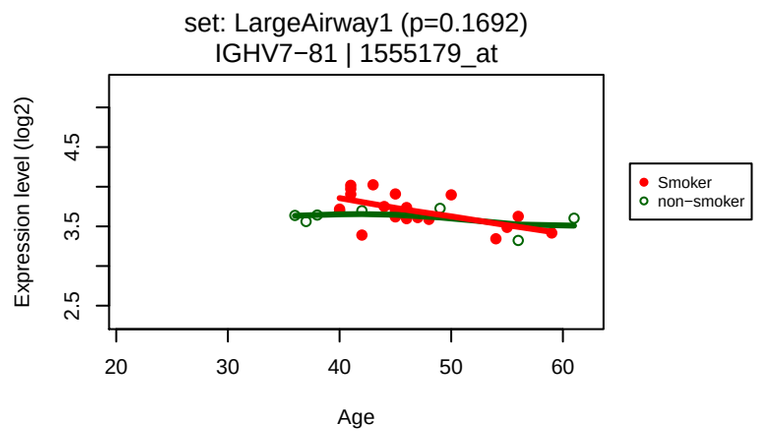

Gene expression of smokers and nonsmokers
gene: IGHV7−81 | 1555179_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2661
p=0.3474
p=0.7048
p=0.1692
p=0.4091
p=0.2973
p=0.3719
p=0.4284
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

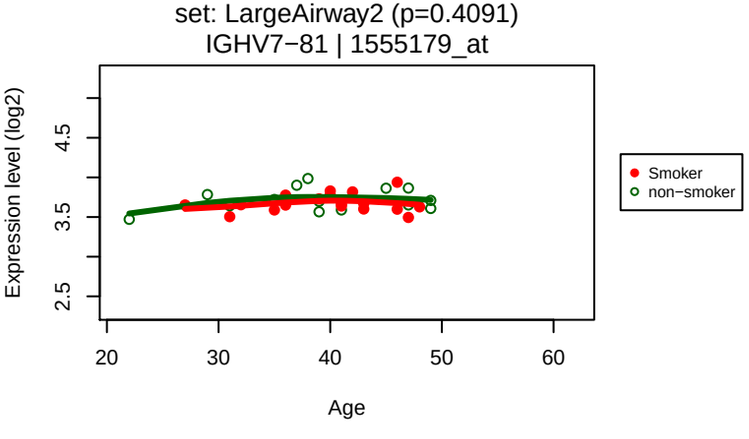

Gene expression ratio (smokers/nonsmokers)
gene: IGHV7−81 | 1555179_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2661
p=0.3474
p=0.7048
p=0.1692
p=0.4091
p=0.2973
p=0.3719
p=0.4284
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

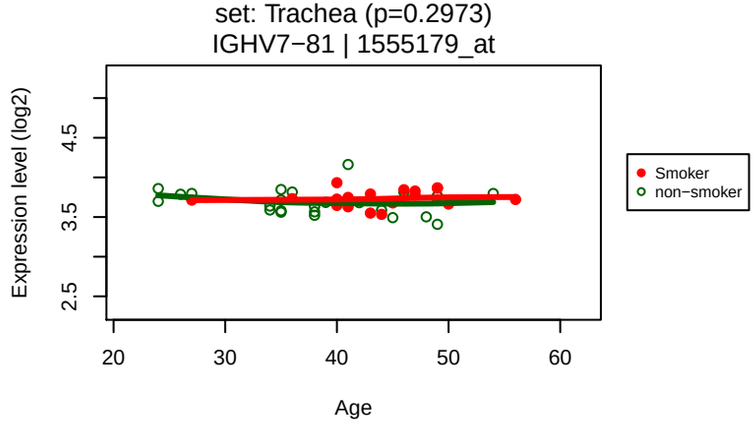

Gene expression ratio in Males
gene: IGHV7−81 | 1555179_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.346
p=0.2521
p=0.0144
p=0.3151
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGHV7−81 | 1555179_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9853
p=0.8992
p=0.1076
p=0.9889
SmallAirway2
LargeAirway2
Trachea
Macrophage2

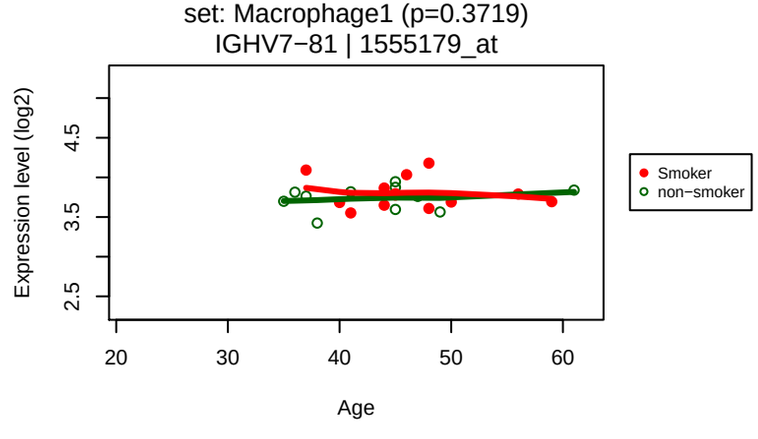

set: SmallAirway1 (p=0.2661)
IGHV7−81 | 1555179_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker

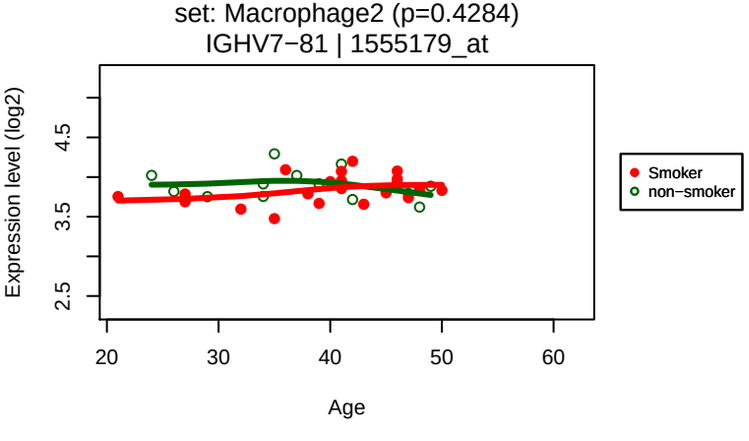

set: SmallAirway2 (p=0.3474)
IGHV7−81 | 1555179_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker

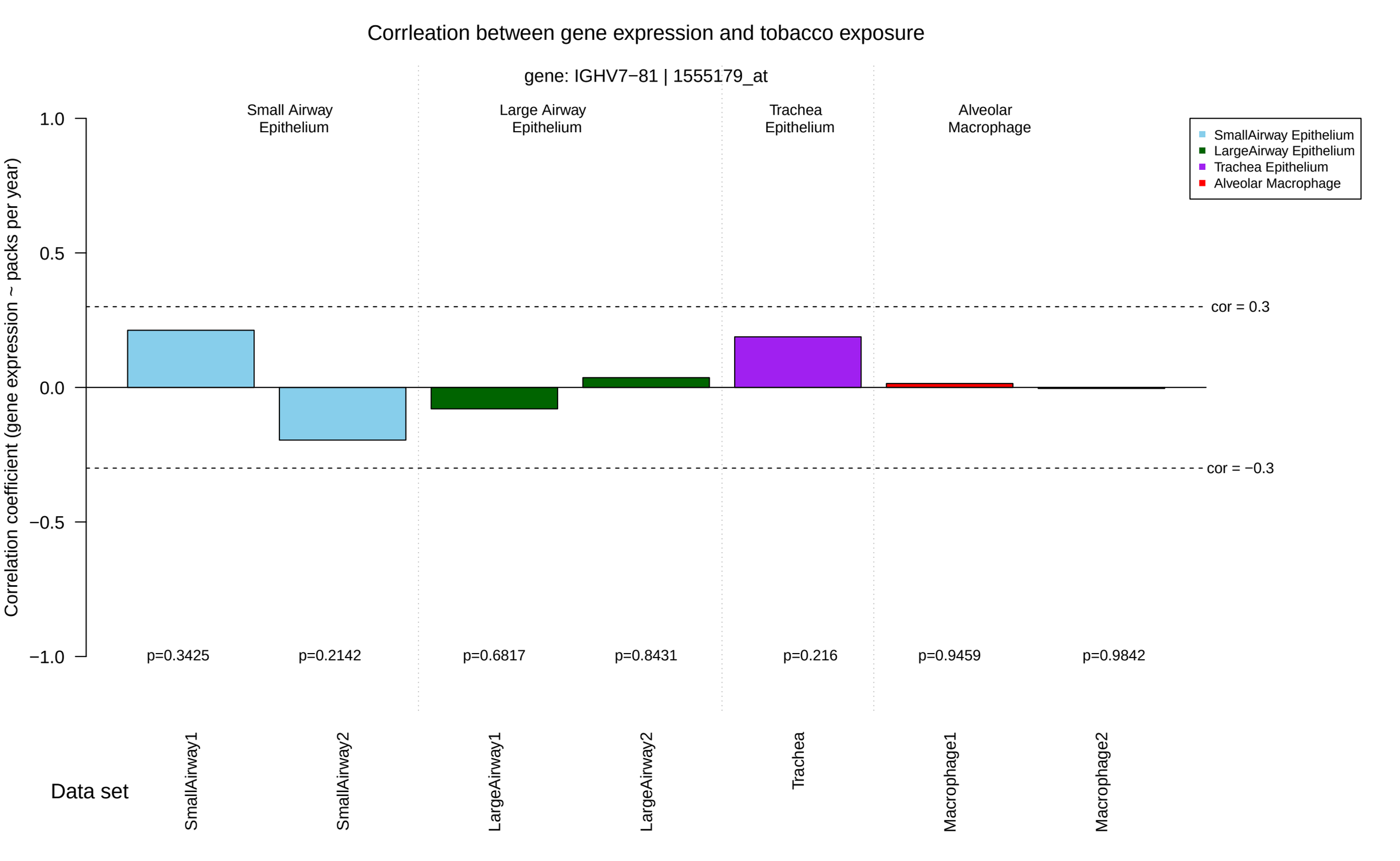

set: LargeAirway1 (p=0.1692)
IGHV7−81 | 1555179_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4091)
IGHV7−81 | 1555179_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.2973)
IGHV7−81 | 1555179_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3719)
IGHV7−81 | 1555179_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4284)
IGHV7−81 | 1555179_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHV7−81 | 1555179_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3425
p=0.2142
p=0.6817
p=0.8431
p=0.216
p=0.9459
p=0.9842
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2